
DESIGN DATA BOOK — SCREWED JOINTS & FASTENERS

Standard Engineering Design Data, Formulas & Solved Protocols (IS: 4218 - 1976)

Fastener Joint Classification & Design Index

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SECTION 1: Standard Tensile Capacity Analysis (M 30 Fasteners)

System Parameters	Design Protocol
Nominal Bolt Diameter: $d = 30 \text{ mm}$ (M 30 Coarse Series)	Retrieve core stress area A_c from Table 11.1
Safe Tensile Stress: $\sigma_t = 42 \text{ MPa} = 42\text{N/mm}^2$	Calculate safe tensile load capacity $P = A_c \sigma_t$
Preload $F_i = 0$	

1. Core Stress Area Lookup (A_c)

Table 11.1 lookup for M 30 coarse series thread

$$\begin{aligned} A_c &= f(d) \\ &= 561\text{mm}^2 \end{aligned}$$

2. Safe Tensile Load Capacity (P)

Maximum allowable axial tension force without initial tightening

$$\begin{aligned} P &= A_c \times \sigma_t \\ &= 561\text{mm}^2 \times 42\text{N/mm}^2 \\ &= 23562 \text{ N} \\ &= \mathbf{23.562 \text{ kN}} \end{aligned}$$

3. Design Output

Safe Tensile Load $P = 23.562 \text{ kN}$

SECTION 2: Initial Tightening Stress & Preload Analysis (Tap Bolts)

System Parameters	Design Protocol
Nominal Tap Bolt Size: $d = 24$ mm (M 24 Coarse Series)	Lookup core root diameter d_c & stress area from Table 11.1
Clamped Rigid Machine Joint	Calculate initial tightening load $P = 2840d$
Neglect external separation load	Calculate induced tensile stress $\sigma_t = \frac{P}{A_c}$ (and $\frac{P}{A_s}$)

1. Core Root Diameter & Area Lookup (d_c, A_c, A_s)

Table 11.1 lookup for M 24 coarse thread

$$d_c = 20.32 \text{ mm}$$

$$\begin{aligned} A_c &= \frac{\pi}{4} d_c^2 \\ &= \frac{\pi}{4} (20.32)^2 \\ &= 324.3 \text{ mm}^2 \\ &\approx 324 \text{ mm}^2 \end{aligned}$$

$$A_s \text{ (Stress Area)} = 353 \text{ mm}^2$$

2. Initial Tightening Load (P)

Empirical initial tightening tension equation

$$\begin{aligned} P &= 2840 \cdot d \\ &= 2840 \times 24 \\ &= 68160 \text{ N} \end{aligned}$$

3. Induced Tightening Stress (σ_t)

Tensile stress induced in thread root area vs
effective stress area

$$\begin{aligned} \sigma_t &= \frac{P}{A_c} \\ &= \frac{68160}{324} \\ &= 210 \text{ N/mm}^2 \\ &= \mathbf{210 \text{ MPa}} \quad (\text{Standard Root Area}) \end{aligned}$$

$$\begin{aligned} \sigma_{t, \text{alt}} &= \frac{P}{A_s} \\ &= \frac{68160}{353} \\ &= \mathbf{193 \text{ MPa}} \quad (\text{Effective Stress Area}) \end{aligned}$$

4. Design Output

Tightening Stress $\sigma_t = 210$ MPa (Root Area $A_c = 324$ mm²) | $\sigma_t = 193$ MPa (Stress Area $A_s = 353$ mm²)

SECTION 3: Lifting Eye Bolt Sizing & Thread Selection

System Parameters

Lifting Load: $P = 60 \text{ kN} = 60000 \text{ N}$

Permissible Tensile Stress: $\sigma_t = 100 \text{ MPa} = 100 \text{ N/mm}^2$

Thread Series: ISO Metric Coarse

Design Protocol

Equate lifting load to tensile capacity $P = \frac{\pi}{4} d_c^2 \sigma_t$

Solve required core root diameter d_c

Select standard bolt from Table 11.1

1. Required Core Root Diameter (d_c)

Axial tension load equation for lifting eye bolt

$$\begin{aligned} P &= \frac{\pi}{4} (d_c)^2 \cdot \sigma_t \\ 60000 &= \frac{\pi}{4} (d_c)^2 \times 100 \\ &= 78.55 (d_c)^2 \\ (d_c)^2 &= \frac{60000}{78.55} \\ &= 763.84 \\ d_c &= \sqrt{763.84} \\ &= 27.64 \text{ mm} \end{aligned}$$

2. Standard Thread Selection (Table 11.1)

Select standard coarse series thread with $d_c \geq 27.64$ mm

$$d_{c, \text{std}} \geq d_c$$

$$d_c = 28.706 \text{ mm}$$

$$d = 33 \text{ mm}$$

3. Design Output

Select Fastener Designation: M 33 Bolt (d = 33 mm, d_c = 28.706 mm)

SECTION 3: Lifting Eye Bolt — MECHANICAL DIAGRAM

11.13 Stress due to Combined Forces

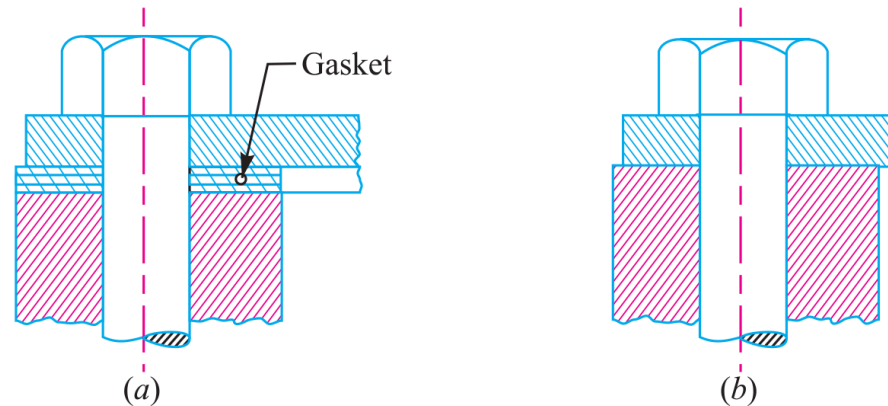


Fig. 11.23

Figure 11.23: Structural Lifting Eye Bolt Configuration and Load Path

SECTION 4: Flanged Shaft Coupling Fastener Sizing under Torsional Shear

System Parameters	Design Protocol
Transmitted Torque: $T = 25 \text{ N} \cdot \text{m} = 25000 \text{ N} \cdot \text{mm}$	Calculate total tangential shearing load $P_s = \frac{T}{R_p}$
Pitch Circle Radius: $R_p = 30 \text{ mm}$	Equate P_s to shear capacity of n bolts $P_s = n(\frac{\pi}{4}d_c^2)\tau$
Bolt Count: $n = 4$ Allowable Shear Stress: $\tau = 30 \text{ MPa} = 30\text{N/mm}^2$	Select standard bolt size from Table 11.1

1. Total Shearing Load (P_s)

Total tangential shearing force acting at PCD

$$\begin{aligned} P_s &= \frac{T}{R_p} \\ &= \frac{25000}{30} \\ &= 833.3 \text{ N} \end{aligned}$$

2. Required Core Diameter (d_c)

Solve root diameter for $n = 4$ bolts carrying shear

$$P_s = n \cdot \left[\frac{\pi}{4} (d_c)^2 \right] \cdot \tau$$

$$833.3 = 4 \cdot \left[\frac{\pi}{4} (d_c)^2 \right] \times 30$$

$$= 94.26 (d_c)^2$$

$$(d_c)^2 = \frac{833.3}{94.26}$$

$$= 8.84$$

$$d_c = \sqrt{8.84}$$

$$= 2.97 \text{ mm}$$

3. Standard Thread Selection (Table 11.1)

Select standard coarse series thread with $d_c \geq$
2.97 mm

$$d_{c, \text{std}} \geq d_c$$

$$d_c = 3.141 \text{ mm}$$

$$d = 4 \text{ mm}$$

4. Design Output

Select Fastener Designation: M 4 Bolt ($d = 4 \text{ mm}$, $d_c = 3.141 \text{ mm}$)

SECTION 5: Safety Valve Fulcrum Pin Sizing & Lever Equilibrium

System Parameters	Design Protocol
Valve Diameter: $D = 100 \text{ mm}$	Calculate upward steam force $F = \frac{\pi}{4} D^2 p$
Blow-off Steam Pressure: $p = 1.6 \text{ N/mm}^2$	Apply lever moment equilibrium for end load W and
Leverage Ratio: $\frac{L_1}{L_2} = 8$ Permissible Tensile Stress: $\sigma_t = 50 \text{ MPa}$	fulcrum load $P = F - W$
	Size fine metric thread d_c for fulcrum pin

1. Total Upward Force on Valve (F)

Total steam pressure load on valve disc

$$\begin{aligned} F &= \frac{\pi}{4} D^2 \cdot p \\ &= \frac{\pi}{4} (100)^2 \times 1.6 \\ &= 12568 \text{ N} \end{aligned}$$

2. Lever End Load (W) & Fulcrum Load (P)

Static moment equilibrium for Type 1 lever

$$W = \frac{F}{\frac{L_1}{L_2}}$$

$$= \frac{12568}{8}$$

$$= 1571 \text{ N}$$

$$P = F - W$$

$$= 12568 - 1571$$

$$= 10997 \text{ N}$$

3. Required Core Diameter (d_c)

Solve root core diameter for fulcrum screw

$$P = \frac{\pi}{4}(d_c)^2 \cdot \sigma_t$$

$$10997 = \frac{\pi}{4}(d_c)^2 \times 50$$

$$= 39.27(d_c)^2$$

$$(d_c)^2 = \frac{10997}{39.27}$$

$$= 280$$

$$d_c = \sqrt{280}$$

$$= 16.7 \text{ mm}$$

4. Standard Fine Thread Selection (Table 11.1)

Select fine series metric thread with $d_c \geq$

16.7 mm

$$d_{c, \text{std}} \geq d_c$$

$$d_c = 18.376 \text{ mm}$$

Fine thread: M 20 x 1.5

5. Design Output

Select Fine Series Thread: M 20 x 1.5 (d = 20 mm, d_c = 18.376 mm)

SECTION 6: Cylinder Head Cover Stud Layout & Leak-Proof Pitch Verification

System Parameters	Design Protocol
Cylinder Effective Diameter: $D = 350 \text{ mm}$	Calculate total upward steam load $P = \frac{\pi}{4} D^2 p$
Maximum Steam Pressure: $p = 1.25 \text{ N/mm}^2$	Assume M 24 studs ($d_c = 20.32 \text{ mm}$) to find count
Permissible Stud Stress: $\sigma_t = 33 \text{ MPa} = 33 \text{ N/mm}^2$	$n = \frac{P}{P_1}$
	Verify leak-proof pitch limits $20\sqrt{d_1} \leq p_c \leq 30\sqrt{d_1}$

1. Total Upward Steam Load (P)

Total gas pressure load on cylinder cover

$$\begin{aligned} P &= \frac{\pi}{4} D^2 \cdot p \\ &= \frac{\pi}{4} (350)^2 \times 1.25 \\ &= 120265 \text{ N} \end{aligned}$$

2. Capacity per M 24 Stud (P_1) & Stud Count (n)

Resisting force capacity per M 24 stud ($d_c = 20.32$ mm)

$$\begin{aligned} P_1 &= \frac{\pi}{4} (d_c)^2 \cdot \sigma_t \\ &= \frac{\pi}{4} (20.32)^2 \times 33 \\ &= 10700 \text{ N} \end{aligned}$$

$$\begin{aligned} n &= \frac{P}{P_1} \\ &= \frac{120265}{10700} \\ &= 11.24 \Rightarrow n = 12 \text{ studs} \end{aligned}$$

3. Pitch Circle Diameter (D_p) & Circumferential Pitch (p_c)

Flange wall $t = 10$ mm, stud hole $d_1 = 25$ mm

$$\begin{aligned} D_p &= D + 2t + 3d_1 \\ &= 350 + 2(10) + 3(25) \\ &= 445 \text{ mm} \end{aligned}$$

$$\begin{aligned}
 p_c &= \frac{\pi D_p}{n} \\
 &= \frac{\pi \times 445}{12} \\
 &= 116.5 \text{ mm}
 \end{aligned}$$

4. Leak-Proof Pitch Verification

Empirical leak-tight pitch limits (d_1 in mm)

$$\begin{aligned}
 p_{c, \min} &= 20\sqrt{d_1} \\
 &= 20\sqrt{25} \\
 &= 100 \text{ mm}
 \end{aligned}$$

$$\begin{aligned}
 p_{c, \max} &= 30\sqrt{d_1} \\
 &= 30\sqrt{25} \\
 &= 150 \text{ mm}
 \end{aligned}$$

100 mm $\leq$ 116.5 mm $\leq$ 150 mm (**SAFE & SATISFACTORY**)

5. Design Output

Use 12 Studs of M 24 Size (n = 12, M 24 Studs)

SECTION 6: Cylinder Cover Stud Layout — MECHANICAL DIAGRAM

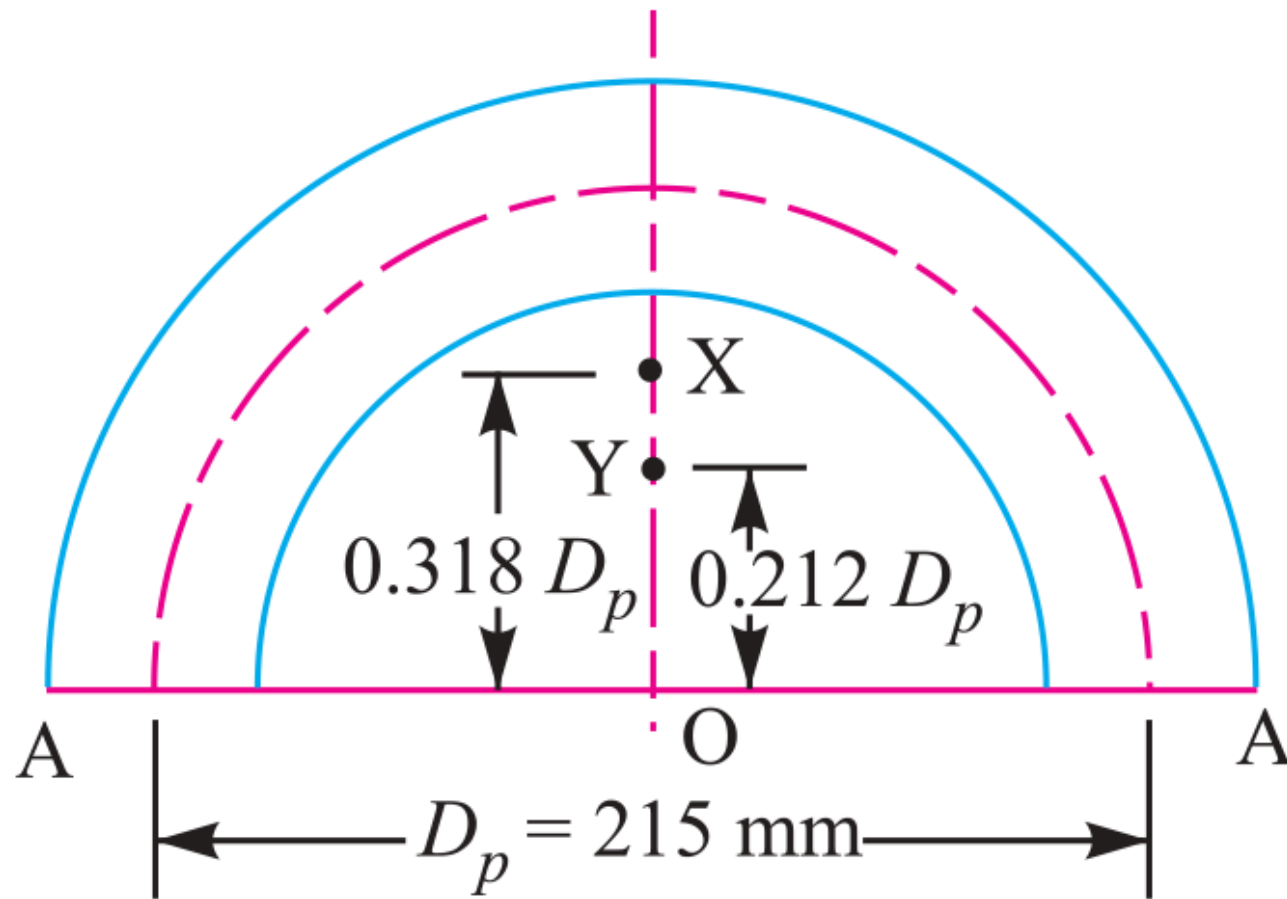


Fig. 11.27

Figure 11.27: Cylinder Head Joint & Stud Pitch Circle Assembly

SECTION 7: Pressure Vessel Inspection Cover Plate & Bolt Sizing

System Parameters	Design Protocol
Inspection Hole Diameter: $D = 120 \text{ mm}$ ($r = 60 \text{ mm}$)	Part 1: Calculate shell wall thickness t via Lamé
Internal Vessel Pressure: $p = 6 \text{ N/mm}^2$	Part 2: Determine bolt count n for M 24 and check pitch p_c
Plate Tensile Stress: $\sigma_t = 60 \text{ MPa}$ Bolt Stress: $\sigma_{tb} = 40 \text{ MPa}$	Part 3: Calculate cover plate thickness t_1 under bending moment

1. Shell Wall Thickness (t)

Lamé equation for thick pressure vessels

$$\begin{aligned} t &= r \left[\sqrt{\frac{\sigma_t + p}{\sigma_t - p}} - 1 \right] \\ &= 60 \left[\sqrt{\frac{60 + 6}{60 - 6}} - 1 \right] \\ &= 60 \left[\sqrt{1.222} - 1 \right] \\ &= 6.3 \text{ mm} \end{aligned}$$

Adopt Shell Thickness $t = 10 \text{ mm}$

2. Fixing Bolts Sizing & Count (n)

Upward pressure load P and M 24 bolt capacity

($d_c = 20.32 \text{ mm}$)

$$\begin{aligned} P &= \frac{\pi}{4} D^2 \cdot p \\ &= \frac{\pi}{4} (120)^2 \times 6 \\ &= 67860 \text{ N} \end{aligned}$$

$$\begin{aligned} P_1 &= \frac{\pi}{4} (d_c)^2 \cdot \sigma_{tb} \\ &= \frac{\pi}{4} (20.32)^2 \times 40 \\ &= 12973 \text{ N} \end{aligned}$$

$$\begin{aligned} n &= \frac{P}{P_1} \\ &= \frac{67860}{12973} \\ &= 5.23 \Rightarrow n = 6 \text{ bolts} \end{aligned}$$

3. PCD & Leak-Proof Pitch Verification

Bolt hole $d_1 = 25$ mm for M 24 bolts

$$\begin{aligned} D_p &= D + 2t + 3d_1 \\ &= 120 + 2(10) + 3(25) \\ &= 215 \text{ mm} \end{aligned}$$

$$\begin{aligned} p_c &= \frac{\pi D_p}{n} \\ &= \frac{\pi \times 215}{6} \\ &= 112.6 \text{ mm} \end{aligned}$$

100 mm $\leq$ 112.6 mm $\leq$ 150 mm (**SAFE & SATISFACTORY**)

4. Cover Plate Thickness (t_1)

Bending moment $M = 0.053PD_p$ and section modulus $Z = \frac{1}{6}wt_1^2$

$$\begin{aligned} M &= 0.053PD_p \\ &= 0.053 \times 67860 \times 215 \\ &= 773265 \text{ N} \cdot \text{mm} \end{aligned}$$

$$\begin{aligned} D_o &= D_p + 3d_1 \\ &= 290 \text{ mm} \end{aligned}$$

$$\begin{aligned}
 w &= D_o - 2d_1 \\
 &= 240 \text{ mm} \\
 Z &= \frac{1}{6}w(t_1)^2 \\
 &= \frac{1}{6} \times 240 \times (t_1)^2 \\
 &= 40(t_1)^2 \\
 60 &= \frac{773265}{40(t_1)^2} \\
 (t_1)^2 &= 322.2 \\
 t_1 &= 17.95 \text{ mm} \\
 &= \mathbf{18 \text{ mm}}
 \end{aligned}$$

5. Design Output

Final Design: 6 Bolts of M 24 Size, Cover Plate Thickness
t₁ = 18 mm

SECTION 8: Preloaded Flange Fasteners with Gasket Compression (Soft Copper)

System Parameters

Steam Pressure: $p = 0.7\text{N/mm}^2$
Number of Bolts: $n = 12$
Effective Cylinder Diameter: $D = 300\text{ mm}$
Allowable Stress: $\sigma_t = 100\text{ MPa}$ | Soft Copper Gasket
Ratio: $K = 0.5$

Design Protocol

Calculate external load per bolt $P_2 = \frac{P_{\text{total}}}{12}$
Formulate resultant axial load $P = 2840d + KP_2$
Solve governing quadratic for nominal diameter d

1. External Pressure Load per Bolt (P_2)

Upward steam pressure load shared by 12 bolts

$$\begin{aligned} P_{\text{total}} &= \frac{\pi}{4} D^2 \cdot p \\ &= \frac{\pi}{4} (300)^2 \times 0.7 \\ &= 49480\text{ N} \end{aligned}$$

$$\begin{aligned}
 P_2 &= \frac{P_{\text{total}}}{n} \\
 &= \frac{49480}{12} \\
 &= 4124 \text{ N}
 \end{aligned}$$

2. Resultant Axial Load per Bolt (P)

Initial tension $P_1 = 2840d$ plus gasket compression load

$$\begin{aligned}
 P &= P_1 + K \cdot P_2 \\
 &= 2840d + 0.5 \times 4124 \\
 &= 2840d + 2062 \text{ N}
 \end{aligned}$$

3. Core Resisting Capacity & Quadratic Equation

Core diameter relation $d_c = 0.84d$

$$\begin{aligned}
 P &= \frac{\pi}{4} (d_c)^2 \cdot \sigma_t \\
 &= \frac{\pi}{4} (0.84d)^2 \times 100 \\
 &= 55.4d^2
 \end{aligned}$$

$$55.4d^2 = 2840d + 2062$$

$$55.4d^2 - 2840d - 2062 = 0$$

$$d^2 - 51.3d - 37.2 = 0$$

4. Quadratic Diameter Solution (d)

Quadratic formula evaluation for nominal bolt diameter

$$\begin{aligned}d &= \frac{51.3 \pm \sqrt{(51.3)^2 - 4(1)(-37.2)}}{2} \\&= \frac{51.3 + \sqrt{2631.69 + 148.8}}{2} \\&= \frac{51.3 + 52.73}{2} \\&= \mathbf{52.01 \text{ mm}}\end{aligned}$$

5. Design Output

Select Fastener Designation: M 52 Bolt ($d = 52 \text{ mm}$)

SECTION 9: Soderberg Fatigue & Preload Design Criterion

System Parameters	Design Protocol
$D = 300 \text{ mm}, p = 1.5\text{N/mm}^2, n = 8, \sigma_y = 330 \text{ MPa}, \sigma_e = 240 \text{ MPa}$	Calculate maximum load $P_1 + K P_2$ & minimum load P_1 per bolt
Initial Preload: $P_1 = 1.5 P_2$, Gasket Factor: $K = 0.5$, “F.S.” = 2	Determine mean load P_m and variable load P_v per bolt
	Apply Soderberg fatigue equation to solve core diameter d_c

1. Total Steam Load (P_2) & Initial Preload (P_1)

Steam pressure force and initial preload tension

$$\begin{aligned} P_2 &= \frac{\pi}{4} D^2 \cdot p \\ &= \frac{\pi}{4} (300)^2 \times 1.5 \\ &= 106040 \text{ N} \end{aligned}$$

$$\begin{aligned}P_1 &= 1.5P_2 \\&= 1.5 \times 106040 \\&= 159060 \text{ N}\end{aligned}$$

2. Maximum & Minimum Load per Bolt

Load allocation shared across $n = 8$ bolts

$$\begin{aligned}P_{\max} &= P_1 + KP_2 \\&= 159060 + 0.5(106040) \\&= 212080 \text{ N} \\P_{\max, \text{ bolt}} &= \frac{212080}{8} \\&= 26510 \text{ N} \\P_{\min, \text{ bolt}} &= \frac{159060}{8} \\&= 19882 \text{ N}\end{aligned}$$

3. Mean (P_m) & Variable (P_v) Loads per Bolt

Mean and alternating cyclic load components

$$\begin{aligned} P_m &= \frac{P_{\text{max, bolt}} + P_{\text{min, bolt}}}{2} \\ &= \frac{26510 + 19882}{2} \\ &= 23196 \text{ N} \end{aligned}$$

$$\begin{aligned} P_v &= \frac{P_{\text{max, bolt}} - P_{\text{min, bolt}}}{2} \\ &= \frac{26510 - 19882}{2} \\ &= 3314 \text{ N} \end{aligned}$$

4. Mean (σ_m) & Variable (σ_v) Stresses

Stress area $A_c = \frac{\pi}{4}(d_c)^2 = 0.7854(d_c)^2$

$$\begin{aligned} \sigma_m &= \frac{P_m}{A_c} \\ &= \frac{23196}{0.7854(d_c)^2} \\ &= \frac{29534}{(d_c)^2} \text{ N/mm}^2 \end{aligned}$$

$$\begin{aligned}\sigma_v &= \frac{P_v}{A_c} \\ &= \frac{3314}{0.7854(d_c)^2} \\ &= \frac{4220}{(d_c)^2} \text{N/mm}^2\end{aligned}$$

5. Soderberg Fatigue Failure Equation

Soderberg line equation substitution

$$\sigma_v = \left(\frac{\sigma_e}{\text{F.S.}} \right) \left[1 - \frac{\sigma_m \cdot \text{F.S.}}{\sigma_y} \right]$$

$$\frac{4220}{(d_c)^2} = \left(\frac{240}{2} \right) \left[1 - \left(\frac{29534}{(d_c)^2} \right) \cdot \left(\frac{2}{330} \right) \right]$$

$$= 120 \left[1 - \frac{179.0}{(d_c)^2} \right]$$

$$\frac{4220 + 21480}{(d_c)^2} = 120$$

$$\frac{25700}{(d_c)^2} = 120$$

$$d_c = \sqrt{214.17}$$

$$= 14.63 \text{ mm}$$

6. Standard Thread Selection (Table 11.1)

Select standard coarse thread with $d_c \geq 14.63 \text{ mm}$

$$d_{c, \text{std}} \geq d_c$$

$$d_c = 14.933 \text{ mm}$$

$$d = 18 \text{ mm}$$

7. Design Output

Select Fastener Designation: M 18 Bolt (d = 18 mm, d_c = 14.933 mm)

SECTION 10: Boiler Longitudinal Bar Stay Sizing

System Parameters	Design Protocol
Pitch of Stays: 350 mm × 350 mm	Calculate area supported by each stay $A = p_s \times p_s$
Steam Pressure: $p = 0.84\text{N/mm}^2$	Calculate steam force $P = A \cdot p$
Permissible Tensile Stress: $\sigma_t = 56 \text{ MPa} = 56\text{N/mm}^2$	Solve required core diameter d_c and select bolt size

1. Area Supported by Each Stay (A)

Area of boiler plate supported per stay

$$\begin{aligned} A &= p_s \times p_s \\ &= 350 \times 350 \\ &= 122500\text{mm}^2 \end{aligned}$$

2. Steam Force Acting on Stay (P)

Total steam pressure load on supported area

$$\begin{aligned} P &= A \cdot p \\ &= 122500\text{mm}^2 \times 0.84\text{N/mm}^2 \\ &= 102900 \text{ N} \end{aligned}$$

3. Required Core Root Diameter (d_c)

Core diameter required for permissible tension stress

$$\begin{aligned} P &= \frac{\pi}{4} (d_c)^2 \cdot \sigma_t \\ 102900 &= \frac{\pi}{4} (d_c)^2 \times 56 \\ &= 43.98 (d_c)^2 \\ (d_c)^2 &= \frac{102900}{43.98} \\ &= 2339.6 \\ d_c &= \sqrt{2339.6} \\ &= 48.37 \text{ mm} \end{aligned}$$

4. Standard Thread Selection (Table 11.1)

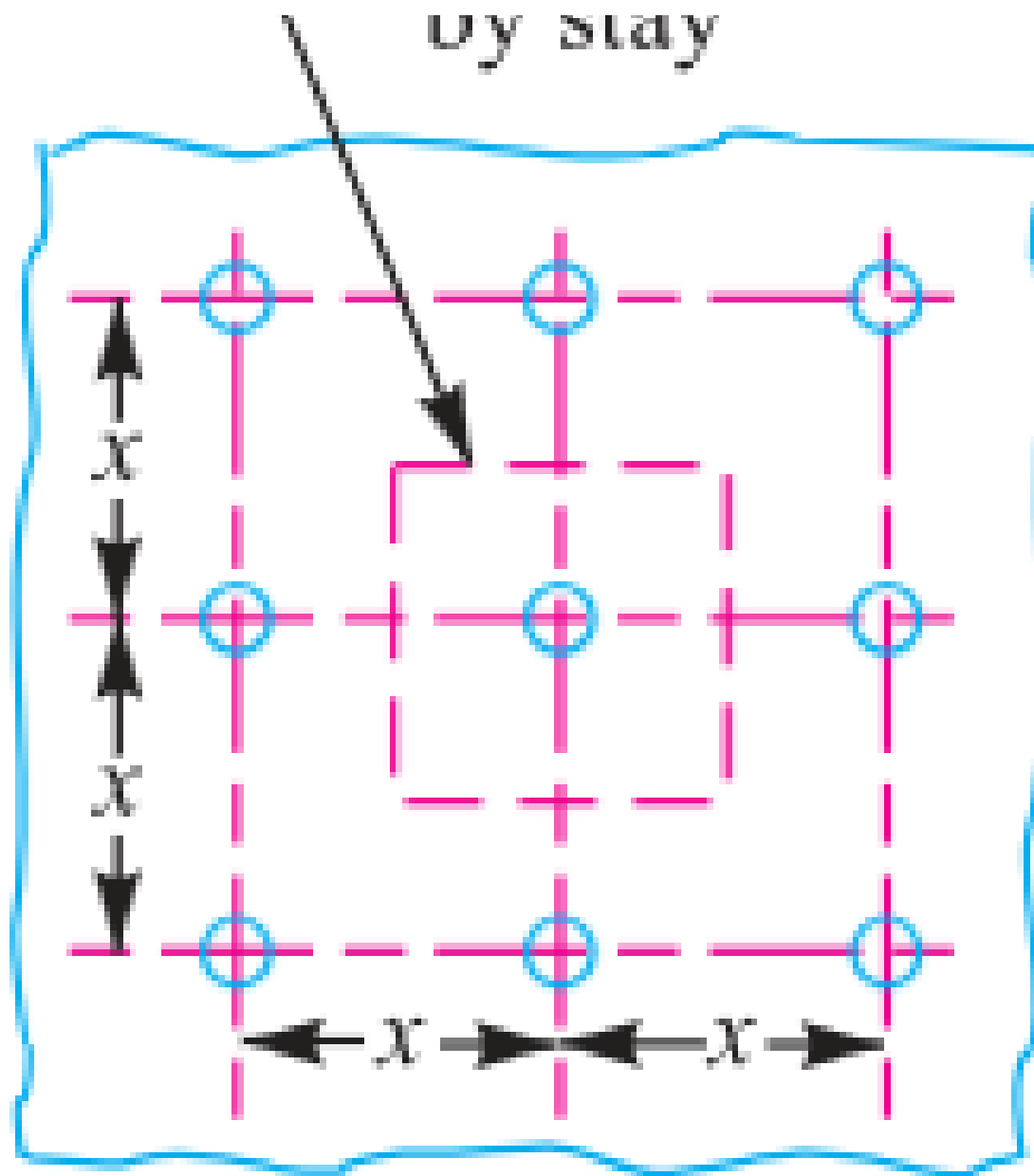
Select standard coarse thread with $d_c \geq 48.37$ mm

$$\begin{aligned} d_{c, \text{std}} &\geq d_c \\ d_c &= 49.177 \text{ mm} \\ d &= 56 \text{ mm} \end{aligned}$$

5. Design Output

Select Fastener Designation: M 56 Bolt (d = 56 mm, d_c = 49.177 mm)

SECTION 10: Boiler Bar Stay Arrangement — MECHANICAL DIAGRAM



SECTION 11: Turned Shank & Drilled Hole Bolts of Uniform Strength

System Parameters	Design Protocol
Nominal Bolt Size: $D_o = 48$ mm (M 48 Coarse Series)	Lookup core root diameter D_c from Table 11.1
Objective: Determine drilled axial hole diameter D for uniform strength	Equate unthreaded drilled shank area to core root area Solve $D = \sqrt{D_o^2 - D_c^2}$

1. Core Diameter Lookup (D_c)

Table 11.1 lookup for M 48 coarse thread

$$\begin{aligned} D_c &= f(D_o) \\ &= 41.795 \text{ mm} \end{aligned}$$

2. Area Equality Condition for Uniform Strength

Equate unthreaded drilled shank area to root area

$$\begin{aligned} \frac{\pi}{4}(D_o^2 - D^2) &= \frac{\pi}{4}D_c^2 \\ D_o^2 - D^2 &= D_c^2 \end{aligned}$$

3. Drilled Axial Hole Diameter (D)

Solve central axial hole diameter

$$\begin{aligned} D &= \sqrt{D_o^2 - D_c^2} \\ &= \sqrt{(48)^2 - (41.795)^2} \\ &= \sqrt{2304 - 1746.82} \\ &= \sqrt{557.18} \\ &= \mathbf{23.64 \text{ mm}} \end{aligned}$$

4. Design Output

Drill Central Axial Hole of Diameter $D = 23.64 \text{ mm}$

SECTION 11: Uniform Strength Fasteners — MECHANICAL DIAGRAM

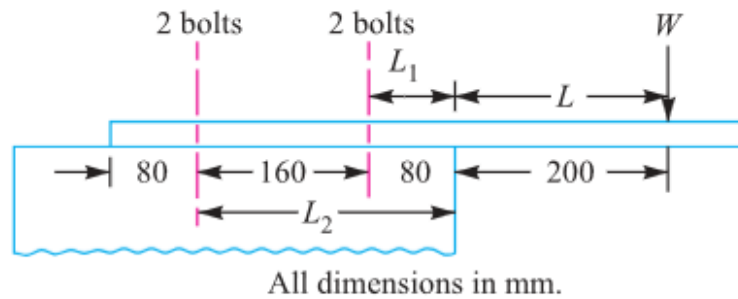


Fig. 11.48

Figure 11.48: Turned Shank Uniform Strength Bolt

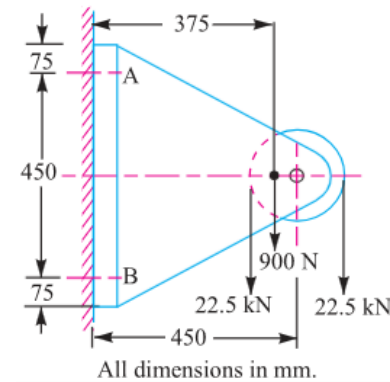


Fig. 11.49

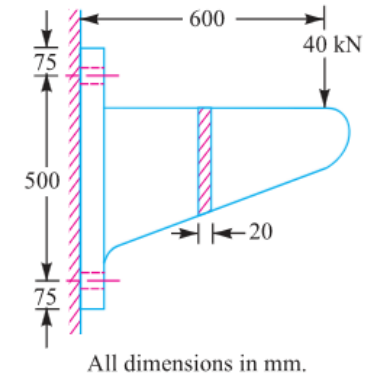


Fig. 11.50

Figure 11.49 & 11.50: Core-Drilled Hole Uniform Strength Bolt

SECTION 12: Wall Brackets under Parallel Eccentric Tensile Loading

System Parameters	Design Protocol
Load: $W = 30 \text{ kN} = 30000 \text{ N}$	Calculate direct tensile load $W_{t1} = \frac{W}{n}$
Eccentricity Arm: $L = 500 \text{ mm}$	Calculate unit secondary load $w = \frac{WL}{2(L_1^2 + L_2^2)}$
Distances: $L_1 = 80 \text{ mm}, L_2 = 250 \text{ mm}$	Calculate total tension $W_t = W_{t1} + wL_2$ and core
Bolt Count: $n = 4$ (2 per row) Allowable Stress: $\sigma_t = 60 \text{ MPa}$	size d_c

1. Direct Tensile Load per Bolt (W_{t1})

Direct load shared equally by 4 bolts

$$\begin{aligned} W_{t1} &= \frac{W}{n} \\ &= \frac{30000}{4} \\ &= 7500 \text{ N} \\ &= 7.5 \text{ kN} \end{aligned}$$

2. Secondary Tensile Load on Upper Bolts

(W_{t2})

Moment equilibrium load factor w

$$\begin{aligned}w &= \frac{W \cdot L}{2(L_1^2 + L_2^2)} \\&= \frac{30 \times 500}{2(80^2 + 250^2)} \\&= \frac{15000}{2(6400 + 62500)} \\&= 0.10886 \text{ kN/mm}\end{aligned}$$

$$\begin{aligned}W_{t2} &= w \cdot L_2 \\&= 0.10886 \times 250 \\&= 27.215 \text{ kN} \\&= 27215 \text{ N}\end{aligned}$$

3. Total Maximum Tensile Load (W_t)

Combined direct and secondary tension on critical top bolts

$$\begin{aligned}W_t &= W_{t1} + W_{t2} \\&= 7.5 + 27.215 \\&= 34.715 \text{ kN} \\&= 34715 \text{ N}\end{aligned}$$

4. Required Core Diameter (d_c)

Solve root diameter for top row fasteners

$$W_t = \frac{\pi}{4}(d_c)^2 \cdot \sigma_t$$

$$34715 = \frac{\pi}{4}(d_c)^2 \times 60$$

$$= 47.12(d_c)^2$$

$$(d_c)^2 = \frac{34715}{47.12}$$

$$= 736.7$$

$$d_c = \sqrt{736.7}$$

$$= 27.14 \text{ mm}$$

5. Standard Thread Selection (Table 11.1)

Select standard coarse thread with $d_c \geq 27.14 \text{ mm}$

$$d_{c, \text{std}} \geq d_c$$

$$d_c = 28.706 \text{ mm}$$

$$d = 33 \text{ mm}$$

6. Design Output

Select Fastener Designation: M 33 Bolt (d = 33 mm, d_c = 28.706 mm)

SECTION 12: Wall Bracket (Parallel Load) — MECHANICAL DIAGRAM

11.19 Eccentric Load Acting Parallel to the Axis of Bolts

Consider a bracket having a rectangular base bolted to a wall by means of four bolts as shown in Fig. 11.31. A little consideration will show that each bolt is subjected to a direct tensile load of

$$W_{t1} = \frac{W}{n}, \text{ where } n \text{ is the number of bolts.}$$

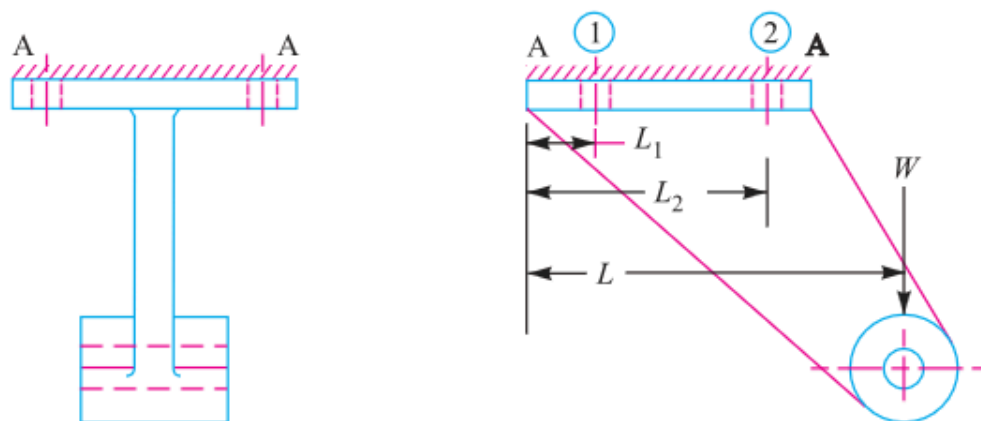


Fig. 11.31. Eccentric load acting parallel to the axis of bolts.

Figure 11.31: Wall Bracket under Parallel Eccentric Load

SECTION 13: Crane Runway T-Bracket Stress Analysis & Fastening Bolts

System Parameters	Design Protocol
Wheel Load: $W = 15 \text{ kN} = 15000 \text{ N}$	Part 1: Calculate centroid $\bar{y}$, moment of inertia I_{GG} , and max arm stresses σ_t, σ_c
Fastening Bolts: $d = 25 \text{ mm}$ ($n = 4$)	
T-Section: Flange 135×25 , Web 175×25 , Total Depth 200 mm	Part 2: Calculate direct & secondary bolt tension to find bolt stress σ_{tb}

1. Centroid ($\bar{y}$) & Section Area (A)

T-section geometry evaluation

$$A_1 = 135 \times 25 = 3375\text{mm}^2$$

$$A_2 = 175 \times 25 = 4375\text{mm}^2$$

$$\begin{aligned} A &= A_1 + A_2 \\ &= 7750\text{mm}^2 \end{aligned}$$

$$\begin{aligned}
 \bar{y} &= \frac{A_1 y_1 + A_2 y_2}{A} \\
 &= \frac{3375 \times 12.5 + 4375 \times 112.5}{7750} \\
 &= 68.95 \text{ mm} \\
 &\approx 69 \text{ mm} \\
 y_1 &= 69 \text{ mm} \quad (\text{Top}) \\
 y_2 &= 200 - 69 = 131 \text{ mm} \quad (\text{Bottom})
 \end{aligned}$$

2. Moment of Inertia (I_{GG}) & Section Moduli (Z_1, Z_2)

Parallel axis theorem for T-section

$$\begin{aligned}
 I_{GG} &= \sum (I_i + A_i d_i^2) \\
 &= \left[\frac{135(25)^3}{12} + 3375(56.5)^2 \right] + \left[\frac{25(175)^3}{12} + 4375(43.5)^2 \right] \\
 &= 30.4 \times 10^6 \text{ mm}^4
 \end{aligned}$$

$$\begin{aligned}
 Z_1 &= \frac{I_{GG}}{y_1} \\
 &= \frac{30.4 \times 10^6}{69} \\
 &= 440.6 \times 10^3 \text{ mm}^3
 \end{aligned}$$

$$\begin{aligned}
 Z_2 &= \frac{I_{GG}}{y_2} \\
 &= \frac{30.4 \times 10^6}{131} \\
 &= 232 \times 10^3 \text{ mm}^3
 \end{aligned}$$

3. Bracket Arm Stresses at Section X-X

Bending moment $M = 15000(200 + 69) = 4.035 \times 10^6 \text{ N} \cdot \text{mm}$

$$\begin{aligned}
 \sigma_{b1} &= \frac{M}{Z_1} \\
 &= \frac{4.035 \times 10^6}{440.6 \times 10^3} \\
 &= 9.16 \text{ N/mm}^2
 \end{aligned}$$

$$\begin{aligned}\sigma_{b2} &= \frac{M}{Z_2} \\ &= \frac{4.035 \times 10^6}{232 \times 10^3} \\ &= 17.4 \text{ N/mm}^2\end{aligned}$$

$$\begin{aligned}\sigma_{t1} &= \frac{W}{A} \\ &= \frac{15000}{7750} \\ &= 1.94 \text{ N/mm}^2\end{aligned}$$

$$\begin{aligned}\sigma_t &= \sigma_{b1} + \sigma_{t1} \\ &= 9.16 + 1.94 \\ &= 11.1 \text{ MPa (Tensile)}\end{aligned}$$

$$\begin{aligned}\sigma_c &= \sigma_{b2} - \sigma_{t1} \\ &= 17.4 - 1.94 \\ &= 15.46 \text{ MPa (Compressive)}\end{aligned}$$

4. Fastening Bolt Loads (W_{t1} , W_{t2} , W_t)

Tilting edge $\mathbb{E}$: $L_1 = 50$ mm, $L_2 = 375$ mm, $L = 525$ mm

$$\begin{aligned}W_{t1} &= \frac{W}{n} \\&= \frac{15000}{4} \\&= 3750 \text{ N}\end{aligned}$$

$$\begin{aligned}w &= \frac{W \cdot L}{2(L_1^2 + L_2^2)} \\&= \frac{15000 \times 525}{2(50^2 + 375^2)} \\&= \frac{7875000}{286250} \\&= 27.5 \text{ N/mm}\end{aligned}$$

$$\begin{aligned}W_{t2} &= w \cdot L_2 \\&= 27.5 \times 375 \\&= 10312.5 \text{ N}\end{aligned}$$

$$\begin{aligned}
 W_t &= W_{t1} + W_{t2} \\
 &= 3750 + 10312.5 \\
 &= 14062.5 \text{ N}
 \end{aligned}$$

5. Maximum Tensile Stress in Fastening Bolts (σ_{tb})

Bolt core diameter $d_c = 0.84d = 0.84 \times 25 = 21 \text{ mm}$

$$\begin{aligned}
 \sigma_{tb} &= \frac{W_t}{\frac{\pi}{4}(d_c)^2} \\
 &= \frac{14062.5}{\frac{\pi}{4}(21)^2} \\
 &= \frac{14062.5}{346.36} \\
 &= 40.6 \text{ MPa}
 \end{aligned}$$

6. Design Output

Arm Tensile $\sigma_t = 11.1 \text{ MPa}$, Arm Compressive $\sigma_c = 15.46 \text{ MPa}$, Bolt Stress $\sigma_{tb} = 40.6 \text{ MPa}$

SECTION 13: Crane Runway T-Bracket — MECHANICAL DIAGRAM

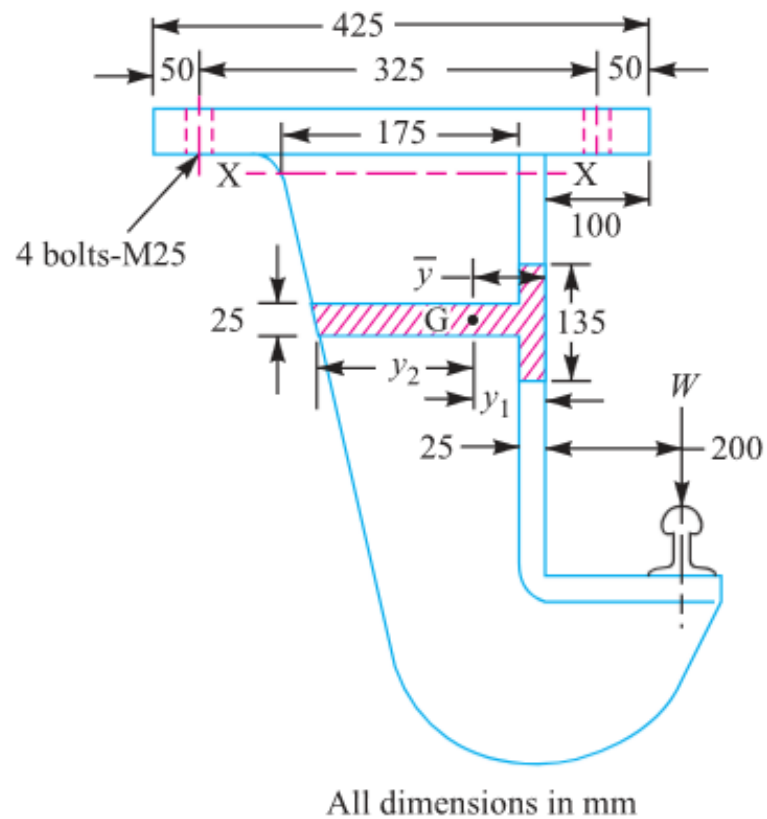


Fig. 11.32

Figure 11.32: Crane Runway T-Bracket Structure

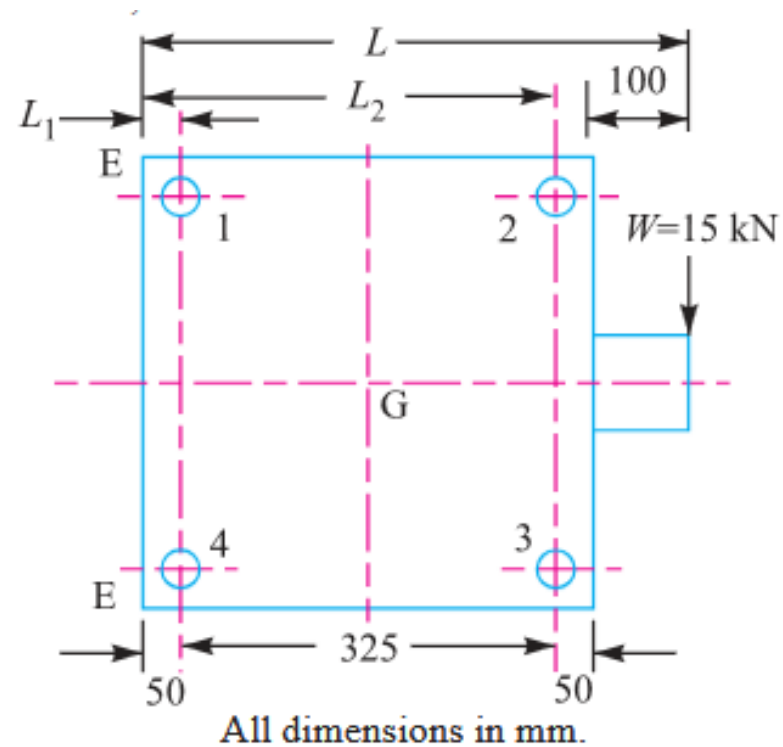


Fig. 11.33

Figure 11.33: T-Section Stress Distribution Diagram

SECTION 14: Travelling Crane Bracket Sizing (Combined Shear & Tension)

System Parameters	Design Protocol
Load: $W = 12 \text{ kN} = 12000 \text{ N}$	Part 1: Calculate direct shear W_s , secondary tension W_t , and equivalent load W_{te} via Maximum Principal Stress Theory
Load Arm: $L = 400 \text{ mm}$	
Distances: $L_1 = 50 \text{ mm}$, $L_2 = 375 \text{ mm}$	
Permissible Tensile Stress: $\sigma_t = 84 \text{ MPa}$ Bolt Count: $n = 4$	Part 2: Calculate arm thickness t for depth $b = 250 \text{ mm}$

1. Direct Shear (W_s) & Secondary Tension (W_t)

Primary shear load and secondary tilting tension per bolt

$$\begin{aligned} W_s &= \frac{W}{n} \\ &= \frac{12000}{4} \\ &= 3000 \text{ N} \\ &= 3 \text{ kN} \end{aligned}$$

$$\begin{aligned}
 W_t &= \frac{W \cdot L \cdot L_2}{2(L_1^2 + L_2^2)} \\
 &= \frac{12 \times 400 \times 375}{2(50^2 + 375^2)} \\
 &= \frac{1800000}{286250} \\
 &= 6.288 \text{ kN} \\
 &= 6288 \text{ N}
 \end{aligned}$$

2. Equivalent Tensile Load (W_{te})

Maximum Principal Stress Theory for combined shear & tension

$$\begin{aligned}
 W_{te} &= \frac{1}{2} \left[W_t + \sqrt{W_t^2 + 4W_s^2} \right] \\
 &= \frac{1}{2} \left[6.29 + \sqrt{(6.29)^2 + 4(3)^2} \right] \\
 &= \frac{1}{2} [6.29 + 8.69] \\
 &= 7.49 \text{ kN} \\
 &= 7490 \text{ N}
 \end{aligned}$$

3. Required Core Diameter (d_c) & Bolt Selection

Table 11.1 lookup for metric coarse thread

$$W_{te} = \frac{\pi}{4} (d_c)^2 \cdot \sigma_t$$

$$7490 = \frac{\pi}{4} (d_c)^2 \times 84$$

$$= 65.97 (d_c)^2$$

$$(d_c)^2 = \frac{7490}{65.97}$$

$$= 113.5$$

$$d_c = \sqrt{113.5}$$

$$= 10.65 \text{ mm}$$

4. Rectangular Cross-Section of Bracket Arm ($t \times b$)

Bending moment $M = W \cdot L = 12 \times 10^3 \times 400 = 4.8 \times 10^6 \text{ N} \cdot \text{mm}$

$$Z = \frac{1}{6} \cdot t \cdot b^2$$

$$\sigma_t = \frac{M}{Z}$$

$$84 = \frac{4.8 \times 10^6}{\frac{1}{6} \cdot t \cdot b^2}$$

$$t \cdot b^2 = \frac{28.8 \times 10^6}{84}$$

$$= 342857 \text{ mm}^3$$

$$t = \frac{342857}{(250)^2}$$

$$= \frac{342857}{62500}$$

$$= 5.5 \text{ mm}$$

5. Design Output

Final Design: M 14 Bolts, Bracket Arm Dimensions 5.5 mm x 250 mm

SECTION 14: Travelling Crane Bracket — MECHANICAL DIAGRAM

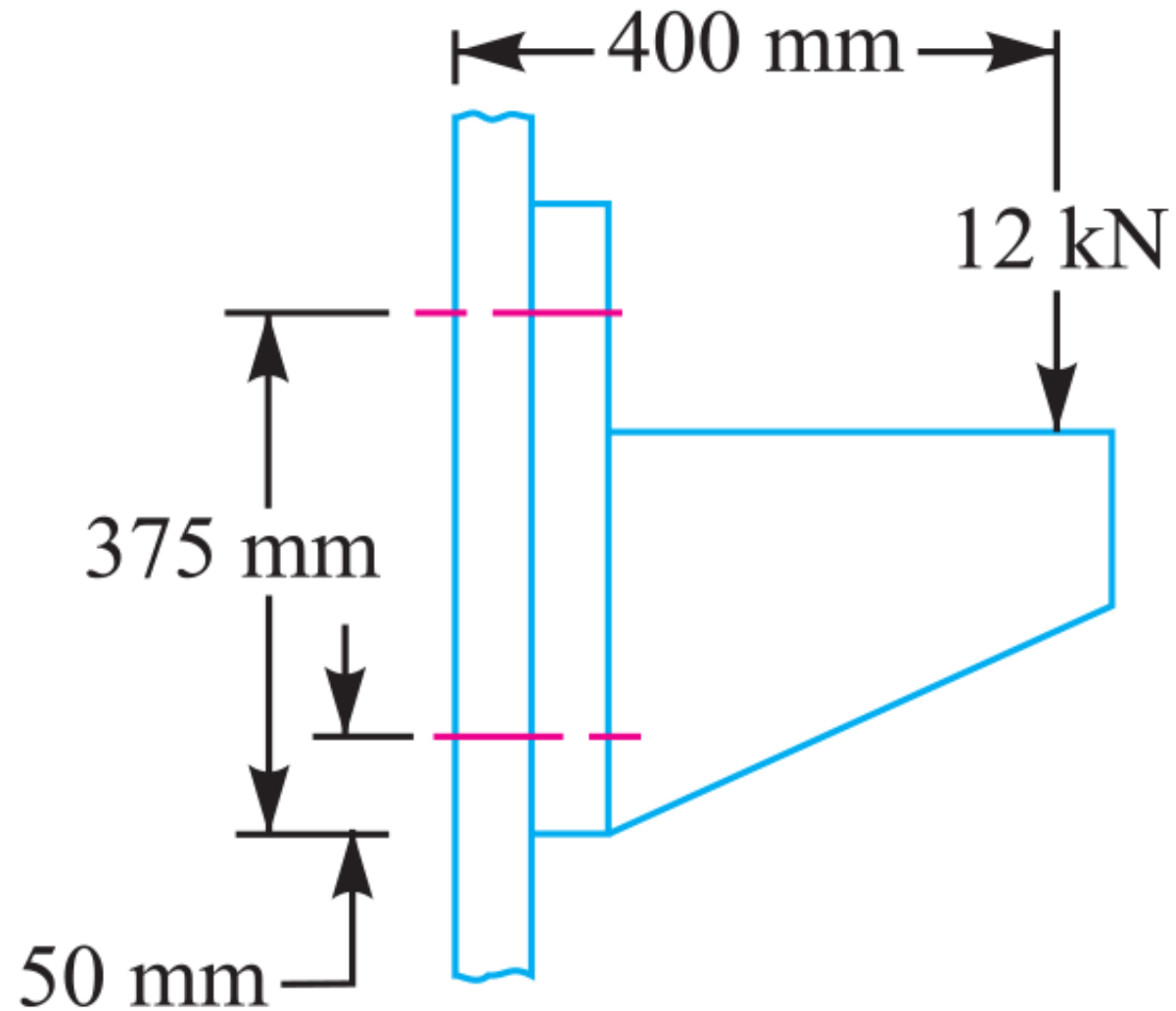


Fig. 11.35

SECTION 15: Inclined Load Bracket Fastener Sizing & Arm Thickness Analysis

System Parameters	Design Protocol
Inclined Load: $W = 40 \text{ kN} = 40000 \text{ N}$ at $\theta = 60^{\text{deg}}$ to vertical	Part 1: Resolve load into W_H , W_V and net moment $T_{\text{net}} = T_V - T_H$
Stresses: $\sigma_t = 70 \text{ MPa}$, $\tau = 50 \text{ MPa}$, $\sigma_c = 105 \text{ MPa}$	Part 2: Calculate bolt tension W_t , equivalent load W_{te} , and size d_c
Arm Depth: $b = 130 \text{ mm}$ Bolts: $n = 4$, $L_1 = 60 \text{ mm}$, $L_2 = 180 \text{ mm}$	Part 3: Calculate arm thickness t via principal tensile stress $\sigma_{t(\text{max})}$

1. Force Resolution & Net Turning Moment (T_{net})

Horizontal and vertical force components

$$\begin{aligned} W_H &= W \sin \theta \\ &= 40 \sin 60^{\text{deg}} \\ &= 34.64 \text{ kN} \\ &= 34640 \text{ N} \end{aligned}$$

$$W_V = W \cos \theta$$

$$= 40 \cos 60^{\text{deg}}$$

$$= 20 \text{ kN}$$

$$= 20000 \text{ N}$$

$$T_H = 34640 \times 20$$

$$= 692.8 \times 10^3 \text{ N} \cdot \text{mm}$$

$$T_V = 20000 \times 175$$

$$= 3500 \times 10^3 \text{ N} \cdot \text{mm}$$

$$T_{\text{net}} = T_V - T_H$$

$$= (3500 - 692.8) \times 10^3$$

$$= 2807.2 \times 10^3 \text{ N} \cdot \text{mm}$$

2. Fixing Bolts Sizing (W_t , W_s , W_{te} , d_c)

Direct & secondary load combination on critical top bolts

$$W_{t1} = \frac{W_H}{n}$$

$$= 8660 \text{ N}$$

$$W_s = \frac{W_V}{n}$$

$$= 5000 \text{ N}$$

$$w = \frac{T_{\text{net}}}{2(L_1^2 + L_2^2)}$$

$$= \frac{2807.2 \times 10^3}{2(60^2 + 180^2)}$$

$$= 38.99 \text{ N/mm}$$

$$\approx 39 \text{ N/mm}$$

$$W_{t2} = w \cdot L_2$$

$$= 39 \times 180$$

$$= 7020 \text{ N}$$

$$W_t = W_{t1} + W_{t2}$$

$$= 8660 + 7020$$

$$= 15680 \text{ N}$$

$$\begin{aligned}
 W_{te} &= \frac{1}{2} \left[W_t + \sqrt{W_t^2 + 4W_s^2} \right] \\
 &= \frac{1}{2} \left[15680 + \sqrt{(15680)^2 + 4(5000)^2} \right] \\
 &= 17140 \text{ N} \\
 (d_c)^2 &= \frac{17140}{55} \\
 &= 311.64 \\
 d_c &= 17.65 \text{ mm}
 \end{aligned}$$

3. Bracket Arm Thickness (t via Principal Stress)

Stresses in upper arm fibre for $b = 130 \text{ mm}$ ($Z = 2817t$)

$$\begin{aligned}
 \sigma_{t1} &= \frac{W_H}{A} \\
 &= \frac{34640}{130t} \\
 &= \frac{266.5}{t}
 \end{aligned}$$

$$\begin{aligned}\sigma_{t2} &= \frac{M_H}{Z} \\ &= \frac{34640 \times 35}{2817t} \\ &= \frac{430.4}{t}\end{aligned}$$

$$\begin{aligned}\sigma_{t3} &= \frac{M_V}{Z} \\ &= \frac{20000 \times 200}{2817t} \\ &= \frac{1420}{t}\end{aligned}$$

$$\begin{aligned}\sigma_t &= \sigma_{t1} + \sigma_{t2} + \sigma_{t3} \\ &= \frac{266.5 + 430.4 + 1420}{t} \\ &= \frac{2116.9}{t}\end{aligned}$$

$$\begin{aligned}\tau &= \frac{W_V}{A} \\ &= \frac{20000}{130t} \\ &= \frac{154}{t}\end{aligned}$$

$$\begin{aligned}\sigma_{t(\max)} &= \frac{\sigma_t}{2} + \frac{1}{2}\sqrt{\sigma_t^2 + 4\tau^2} \\ &= \frac{1058.45}{t} + \frac{1069.6}{t} \\ &= \frac{2128.05}{t} \\ 70 &= \frac{2128.05}{t} \\ t &= 30.4 \text{ mm} \\ &= \mathbf{31 \text{ mm}}\end{aligned}$$

4. Design Output

Final Design: M 22 Bolts, Bracket Arm Thickness t = 31 mm

SECTION 15: Inclined Load Bracket — MECHANICAL DIAGRAM

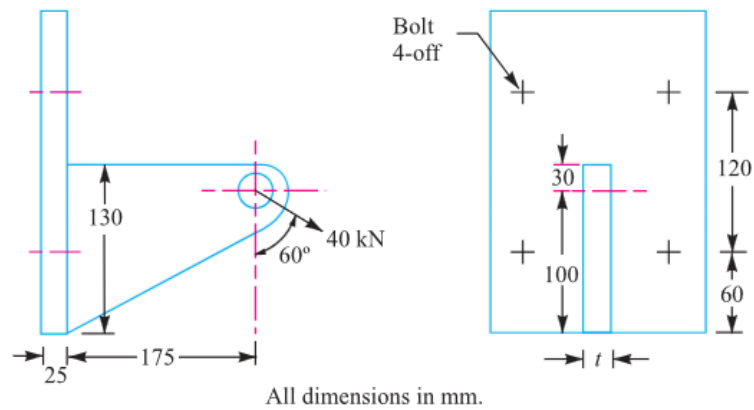


Fig 11.36

Figure 11.36: Wall Bracket under Inclined Load

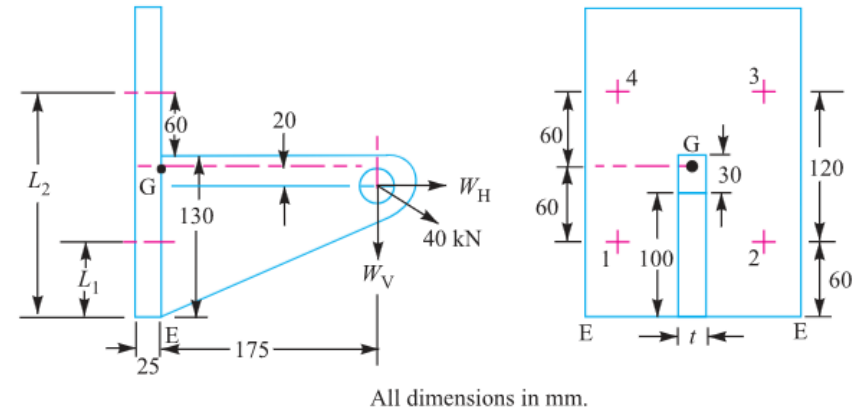


Fig. 11.37

Figure 11.37: Load Resolution & Moment Arm Diagram

SECTION 16: Offset I-Section Bracket under Inclined Loading

System Parameters	Design Protocol
Pull Load: $W = 10 \text{ kN} = 10000 \text{ N}$ at $\theta = 60^{\text{deg}}$ to vertical	Part 1: Resolve load into W_H , W_V and net moment
Stresses: $\sigma_t = 100 \text{ MPa}$, $\tau = 60 \text{ MPa}$	$T_{\text{net}} = T_V - T_H$
Bolt Count: $n = 4$ I-Section Flange Width: $b = 3t$	Part 2: Calculate bolt tension W_t , equivalent load W_{te} , and size d_c
	Part 3: Calculate I-section arm thickness t and width $b = 3t$

1. Force Resolution & Net Turning Moment (T_{net})

Component forces and net moment about tilting edge E

$$\begin{aligned} W_H &= W \sin \theta \\ &= 10 \sin 60^{\text{deg}} \\ &= 8.66 \text{ kN} \\ &= 8660 \text{ N} \end{aligned}$$

$$W_V = W \cos \theta$$

$$= 10 \cos 60^{\text{deg}}$$

$$= 5 \text{ kN}$$

$$= 5000 \text{ N}$$

$$T_H = 433 \text{ N} \cdot \text{m} \quad (\text{Clockwise})$$

$$T_V = 1500 \text{ N} \cdot \text{m} \quad (\text{Counterclockwise})$$

$$T_{\text{net}} = T_V - T_H$$

$$= 1500 - 433$$

$$= 1067 \text{ N} \cdot \text{m}$$

2. Fixing Bolts Sizing (W_t , W_s , W_{te} , d_c)

Bolt distances $L_1 = 0.0375 \text{ m}$, $L_2 = 0.2125 \text{ m}$

$$W_{t1} = \frac{W_H}{n}$$

$$= 2165 \text{ N}$$

$$W_s = \frac{W_V}{n}$$

$$= 1250 \text{ N}$$

$$\begin{aligned}w &= \frac{T_{\text{net}}}{2(L_1^2 + L_2^2)} \\&= \frac{1067}{2(0.0375^2 + 0.2125^2)} \\&= 11470 \text{ N/m}\end{aligned}$$

$$\begin{aligned}W_{t2} &= w \cdot L_2 \\&= 11470 \times 0.2125 \\&= 2435 \text{ N}\end{aligned}$$

$$\begin{aligned}W_t &= W_{t1} + W_{t2} \\&= 2165 + 2435 \\&= 4600 \text{ N}\end{aligned}$$

$$\begin{aligned}
 W_{te} &= \frac{1}{2} \left[W_t + \sqrt{W_t^2 + 4W_s^2} \right] \\
 &= \frac{1}{2} \left[4600 + \sqrt{(4600)^2 + 4(1250)^2} \right] \\
 &= 4920 \text{ N} \\
 (d_c)^2 &= \frac{4920}{78.55} \\
 &= 62.63 \\
 d_c &= 7.91 \text{ mm}
 \end{aligned}$$

3. Dimensions of I-Section Bracket Arm (t and $b = 3t$)

Cross-section area $A = 9t^2$, section modulus $Z = 10.7t^3$

$$\begin{aligned}
 \sigma_{t1} &= \frac{W_H}{A} \\
 &= \frac{8660}{9t^2} \\
 &= \frac{962}{t^2}
 \end{aligned}$$

$$\begin{aligned}\sigma_{t2} &= \frac{M_H}{Z} \\ &= \frac{433 \times 10^3}{10.7t^3} \\ &= \frac{40.5 \times 10^3}{t^3}\end{aligned}$$

$$\begin{aligned}\sigma_{t3} &= \frac{M_V}{Z} \\ &= \frac{1500 \times 10^3}{10.7t^3} \\ &= \frac{140.2 \times 10^3}{t^3}\end{aligned}$$

$$\begin{aligned}\sigma_t &= \sigma_{t1} - \sigma_{t2} + \sigma_{t3} \\ &= \frac{962}{t^2} + \frac{99.7 \times 10^3}{t^3}\end{aligned}$$

$$100 = \frac{962}{t^2} + \frac{99.7 \times 10^3}{t^3} \Rightarrow t = 10.4 \text{ mm}$$

$$b = 3t = 31.2 \text{ mm}$$

4. Design Output

Final Design: M 10 Bolts, I-Section Arm Dimensions $t = 10.4$ mm, $b = 31.2$ mm

SECTION 17: Pillar Crane Circular Base Foundation Flange Sizing

System Parameters	Design Protocol
Bolt Count: $n = 8$	Calculate distance from tilting edge $A - A$: $L = e - R$
Bolt Circle Diameter: $d = 1.6 \text{ m} \Rightarrow r = 0.8 \text{ m}$	
Pillar Base Diameter: $D = 2 \text{ m} \Rightarrow R = 1 \text{ m}$	Calculate maximum tensile load W_t on critical bolt using PCD formula
Crane Load: $W = 100 \text{ kN} = 100000 \text{ N}$ at distance $e = 5 \text{ m}$ Allowable Stress: $\sigma_t = 100 \text{ MPa}$	Solve required core root diameter d_c and select bolt size

1. Tilting Arm Distance (L)

Distance measured from base fulcrum edge $A - A$ of radius $R = 1 \text{ m}$

$$\begin{aligned} L &= e - R \\ &= 5 - 1 \\ &= 4 \text{ m} \end{aligned}$$

2. Maximum Tensile Load on Critical Bolt (W_t)

Circular foundation bolt layout equation for tilting about base edge

$$\begin{aligned} W_t &= \frac{2 \cdot W \cdot L \cdot (R + r)}{n(2R^2 + r^2)} \\ &= \frac{2 \times (100000) \times 4 \times (1 + 0.8)}{8 \times (2(1)^2 + 0.8^2)} \\ &= \frac{800000 \times 1.8}{8 \times 2.64} \\ &= \frac{1440000}{21.12} \\ &= 68.18 \times 10^3 \text{ N} \\ &= 68180 \text{ N} \end{aligned}$$

3. Required Core Root Diameter (d_c)

Solve root diameter for allowable stress

$$\begin{aligned}W_t &= \frac{\pi}{4}(d_c)^2 \cdot \sigma_t \\68180 &= \frac{\pi}{4}(d_c)^2 \times 100 \\&= 78.54(d_c)^2 \\(d_c)^2 &= \frac{68180}{78.54} \\&= 868.1 \\d_c &= \sqrt{868.1} \\&= 29.46 \text{ mm}\end{aligned}$$

4. Standard Thread Selection (Table 11.1)

Select standard coarse thread with $d_c \geq 29.46 \text{ mm}$

$$\begin{aligned}d_{c, \text{std}} &\geq d_c \\d_c &= 31.093 \text{ mm} \\d &= 36 \text{ mm}\end{aligned}$$

5. Design Output

Select Fastener Designation: M 36 Bolt (d = 36 mm, d_c = 31.093 mm)

SECTION 17: Pillar Crane Foundation Flange — MECHANICAL DIAGRAM

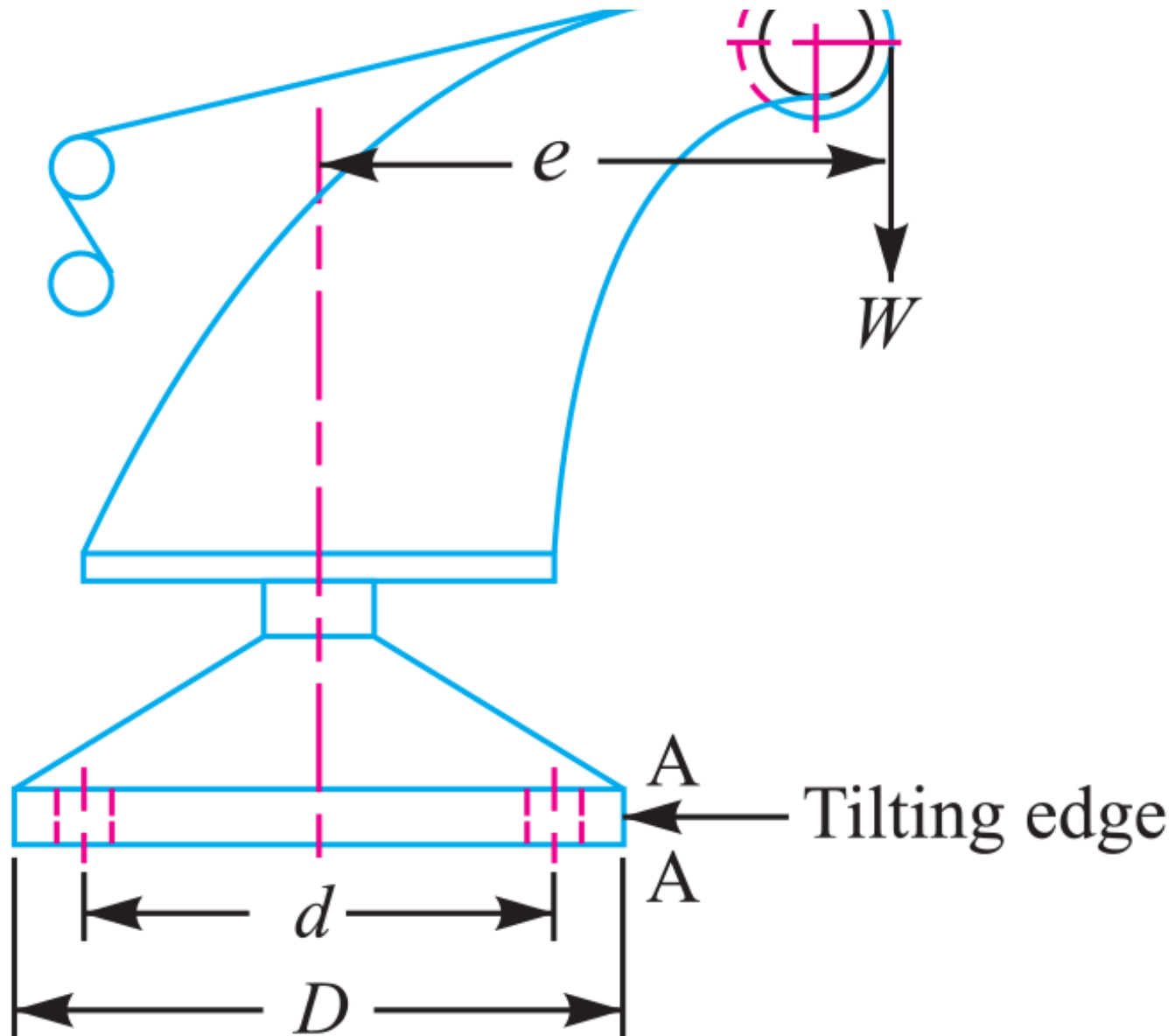


Fig 11.42

SECTION 18: Circular Flanged Bearing Base Fastener Sizing

System Parameters	Design Protocol
Bolt Count: $n = 4$	For $n = 4$ bolts on circular flange, critical bolt lies at
Bolt Circle Diameter: $d = 500 \text{ mm} \Rightarrow r = 250 \text{ mm}$	angle $\alpha = \frac{180^{\text{deg}}}{n} = 45^{\text{deg}}$
Bearing Flange Diameter: $D = 650 \text{ mm} \Rightarrow R = 325 \text{ mm}$	Apply circular PCD overturning equation for W_t
Overturning Load: $W = 400 \text{ kN} = 400000 \text{ N}$ at arm $L = 250 \text{ mm}$ Stress: $\sigma_t = 60 \text{ MPa}$	Solve required core root diameter d_c

1. Maximum Tensile Load on Critical Bolt (W_t)

Overturning equation for $n = 4$ bolts at $\alpha = 45^{\text{deg}}$

$$\begin{aligned} W_t &= \frac{2 \cdot W \cdot L \cdot \left[R + r \cos\left(\frac{180^{\text{deg}}}{n}\right) \right]}{n(2R^2 + r^2)} \\ &= \frac{2 \times (400000) \times 250 \times [325 + 250 \cos 45^{\text{deg}}]}{4 \times (2(325)^2 + (250)^2)} \\ &= \frac{200 \times 10^6 \times [325 + 176.78]}{4 \times 273750} \\ &= \frac{200 \times 10^6 \times 501.78}{1095000} \\ &= 91643 \text{ N} \end{aligned}$$

2. Required Core Root Diameter (d_c)

Solve root diameter for allowable stress

$$W_t = \frac{\pi}{4}(d_c)^2 \cdot \sigma_t$$

$$91643 = \frac{\pi}{4}(d_c)^2 \times 60$$

$$= 47.13(d_c)^2$$

$$(d_c)^2 = \frac{91643}{47.13}$$

$$= 1944.5$$

$$d_c = \sqrt{1944.5}$$

$$= 44.1 \text{ mm}$$

3. Standard Thread Selection (Table 11.1)

Select standard coarse thread with $d_c \geq 44.1 \text{ mm}$

$$d_{c, \text{std}} \geq d_c$$

$$d_c = 45.795 \text{ mm}$$

$$d = 52 \text{ mm}$$

4. Design Output

Select Fastener Designation: M 52 Bolt (d = 52 mm, d_c = 45.795 mm)

SECTION 18: Flanged Bearing Base — MECHANICAL DIAGRAM

11.21 Eccentric Load on a Bracket with Circular Base

Sometimes the base of a bracket is made circular as in case of a flanged bearing of a heavy machine tool and pillar crane etc. Consider a round flange bearing of a machine tool having four bolts as shown in Fig. 11.40.

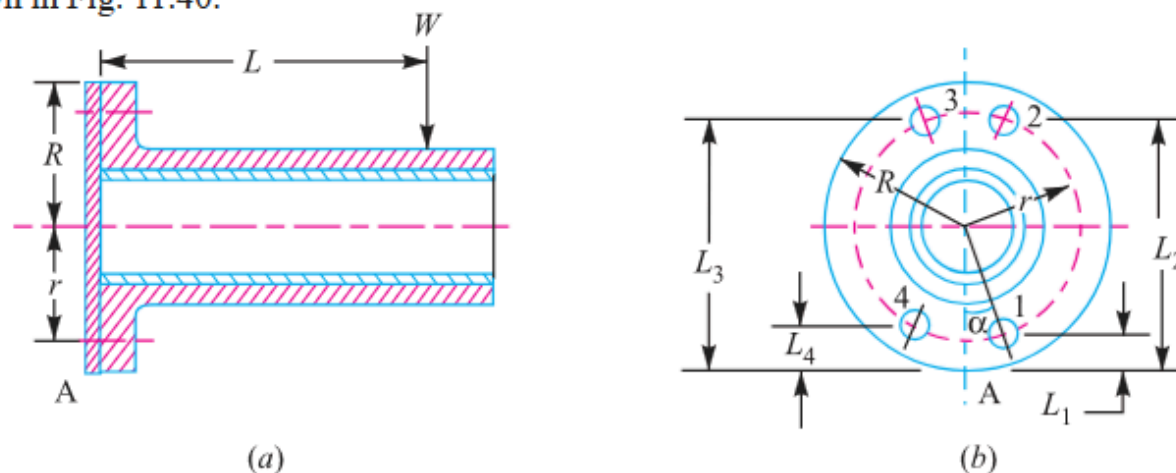


Fig. 11.40. Eccentric load on a bracket with circular base.

Figure 11.40: Circular Flanged Bearing Base PCD Layout

SECTION 19: Pillar Crane Circular Foundation 2-Line Overturning Analysis

System Parameters	Design Protocol
Base Diameter: $D = 600 \text{ mm} \Rightarrow R = 300 \text{ mm} = 0.3 \text{ m}$	Part 1: Analyze line X-X (45^{deg} to bolts) to find max distance e
PCD: $d = 500 \text{ mm} \Rightarrow r = 250 \text{ mm} = 0.25 \text{ m}$	Part 2: Analyze line Y-Y (in line with bolts) at same e
Fasteners: $n = 4 \text{ M } 30 \text{ bolts } (A_c = 561\text{mm}^2)$	to find max induced stress σ_{max}
Overturning Load: $W = 60 \text{ kN} = 60000 \text{ N}$ Allowable	
Stress: $\sigma_t = 60 \text{ MPa}$	

1. Line X-X Analysis (45^{deg} Tilting)

Tensile capacity $P_{\text{cap}} = 561 \times 60 = 33.66 \text{ kN}$,
direct load $W_{\text{t1}} = \frac{60}{4} = 15 \text{ kN}$

$$\begin{aligned} P_{\text{max}} &= P_{\text{cap}} + W_{\text{t1}} \\ &= 33.66 + 15 \\ &= 48.66 \text{ kN} \end{aligned}$$

$$\begin{aligned} L_1 &= R - r \cos 45^{\text{deg}} \\ &= 0.3 - 0.25(0.7071) \\ &= 0.123 \text{ m} \end{aligned}$$

$$L_2 = R + r \cos 45^{\text{deg}}$$

$$= 0.477 \text{ m}$$

$$w = \frac{P_{\text{max}}}{L_2}$$

$$= \frac{48.66}{0.477}$$

$$= 102 \text{ kN/m}$$

$$M_{\text{res}} = 2 \cdot w \cdot [L_1^2 + L_2^2]$$

$$= 2 \times 102 \times [(0.123)^2 + (0.477)^2]$$

$$= 49.4 \text{ kN} \cdot \text{m}$$

2. Maximum Load Distance (e)

Overturning moment $M_{\text{overturn}} = W(e - R) = 60(e - 0.3)$

$$M_{\text{overturn}} = W(e - R)$$

$$60(e - 0.3) = 49.4$$

$$e - 0.3 = \frac{49.4}{60}$$

$$= 0.8233 \text{ m}$$

$$e = 0.8233 + 0.3$$

$$= \mathbf{1.123 \text{ m}}$$

3. Line Y-Y Analysis (In-Line with Bolts)

Distances: $L_1 = 0.05 \text{ m}$, $L_2 = 0.3 \text{ m}$, $L_3 = 0.55 \text{ m}$

$$M_{\text{res}} = w \cdot [L_1^2 + 2L_2^2 + L_3^2]$$

$$= w \cdot [(0.05)^2 + 2(0.3)^2 + (0.55)^2]$$

$$= 0.485 \cdot w$$

$$0.485 \cdot w = 49.4$$

$$w = \frac{49.4}{0.485}$$

$$= 102 \text{ kN/m}$$

4. Line Y-Y Maximum Induced Stress ($\sigma_{\max}$)

Secondary load $W_{t2} = w \cdot L_3 = 102 \times 0.55 = 56.1 \text{ kN}$

$$P_{\text{net}} = W_{t2} - W_{t1}$$

$$= 56.1 - 15$$

$$= 41.1 \text{ kN}$$

$$= 41100 \text{ N}$$

$$\sigma_{\max} = \frac{P_{\text{net}}}{A_c}$$

$$= \frac{41100}{516}$$

$$= 79.65 \text{ MPa}$$

5. Design Output

Final Answers: (1) Maximum Distance $e = 1.123 \text{ m}$, (2) Maximum Induced Stress $\sigma_{\max} = 79.65 \text{ MPa}$

SECTION 19: Pillar Crane 2-Line Analysis — MECHANICAL DIAGRAM

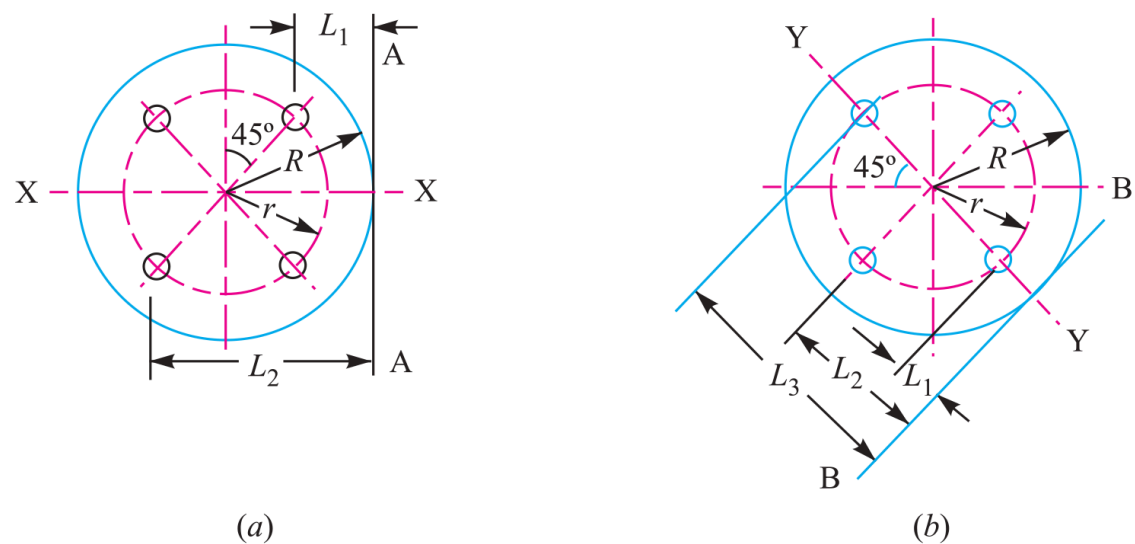


Fig. 11.43

Figure 11.43: 2-Line Overturning Axis Analysis (X-X vs Y-Y)

SECTION 20: Solid Forged Bracket under Combined Bending, Torsion & Coplanar Shear

System Parameters	Design Protocol
Vertical Load: $W = 13.5 \text{ kN} = 13500 \text{ N}$	Part 1: Size arm dia D for combined bending (M) and torsion (T)
Eccentricity: $e = 250 \text{ mm}$	Part 2: Size arm dia d for pure bending (M)
Stresses: Tensile $\sigma_t = 110 \text{ MPa}$, Shear $\tau = 65 \text{ MPa}$	Part 3: Calculate top bolt tensile load W_t
Flange: 4 bolts in square layout	Part 4: Calculate maximum resultant shear force W_s on bolts

1. Diameter D for Arm (Combined Bending + Torsion)

$$M = 13500 \times (300 - 25) = 3.7125 \times 10^6 \text{ N} \cdot \text{mm}$$

$$T = 13500 \times 250 = 3.375 \times 10^6 \text{ N} \cdot \text{mm}$$

$$\begin{aligned} T_e &= \sqrt{M^2 + T^2} \\ &= \sqrt{(3.7125 \times 10^6)^2 + (3.375 \times 10^6)^2} \\ &= 5.017 \times 10^6 \text{ N} \cdot \text{mm} \end{aligned}$$

$$\begin{aligned}T_e &= \frac{\pi}{16} D^3 \cdot \tau \\5.017 \times 10^6 &= \frac{\pi}{16} D^3 \times 65 \\&= 12.76 D^3 \\D^3 &= \frac{5.017 \times 10^6}{12.76} \\&= 393.2 \times 10^3 \\D &= \sqrt[3]{393.2 \times 10^3} \\&= 73.24 \text{ mm} \\&= \mathbf{75 \text{ mm}}\end{aligned}$$

2. Diameter d for Arm (Pure Bending)

Bending moment $M = 13500 \times (250 - 37.5) = 2.8688 \times 10^6 \text{ N} \cdot \text{mm}$

$$\begin{aligned}\sigma_t &= \frac{M}{Z} \\ 110 &= \frac{2.8688 \times 10^6}{\frac{\pi}{32} d^3} \\ &= \frac{2.8688 \times 10^6}{0.0982 d^3} \\ d^3 &= \frac{29.214 \times 10^6}{110} \\ &= 265.58 \times 10^3 \\ d &= \sqrt[3]{265.58 \times 10^3} \\ &= 64.28 \text{ mm} \\ &= \mathbf{65 \text{ mm}}\end{aligned}$$

3. Tensile Load on Each Top Bolt (W_t)

Tilting distances $L_1 = 37.5 \text{ mm}$, $L_2 = 237.5 \text{ mm}$

$$\begin{aligned}M_{\text{res}} &= 2 \cdot w \cdot (L_1^2 + L_2^2) \\ &= 2 \cdot w \cdot (37.5^2 + 237.5^2) \\ &= 115625 \cdot w\end{aligned}$$

$$\begin{aligned}
 M_{\text{overturn}} &= W \cdot L \\
 &= 13500 \times 300 \\
 &= 4.05 \times 10^6 \text{ N} \cdot \text{mm} \\
 w &= \frac{4.05 \times 10^6}{115625} \\
 &= 35.03 \text{ N/mm}
 \end{aligned}$$

$$\begin{aligned}
 W_t &= w \cdot L_2 \\
 &= 35.03 \times 237.5 \\
 &= 8320 \text{ N} \\
 &= \mathbf{8.32 \text{ kN}}
 \end{aligned}$$

4. Maximum Shearing Force on Each Bolt (W_s)

Primary shear $W_{s1} = \frac{13500}{4} = 3375 \text{ N}$, radial dist
 $r_i = \sqrt{100^2 + 100^2} = 141.4 \text{ mm}$

$$\begin{aligned}
 W_{s2} &= \frac{W \cdot e \cdot r_i}{\sum r_i^2} \\
 &= \frac{13500 \times 250 \times 141.4}{4 \times (141.4)^2} \\
 &= 5967 \text{ N}
 \end{aligned}$$

$$W_s = \sqrt{W_{s1}^2 + W_{s2}^2 - 2W_{s1}W_{s2}\cos\theta}$$

$$W_s \quad (\text{Bolts 1 \& 4}) = 4303 \text{ N}$$

$$W_s = \sqrt{W_{s1}^2 + W_{s2}^2 + 2W_{s1}W_{s2}\cos\theta}$$

$$W_s \quad (\text{Bolts 2 \& 3}) = 8687 \text{ N}$$

5. Design Output

Final Answers: (1) D = 75 mm, (2) d = 65 mm, (3) Top Bolt W_t = 8.32 kN, (4) Max Shear Force = 8687 N

SECTION 20: Solid Forged Bracket — MECHANICAL DIAGRAM

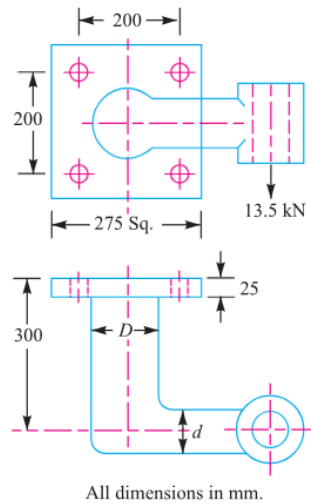


Fig. 11.45

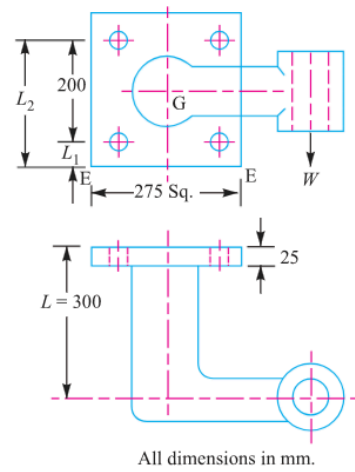


Fig. 11.46

Figure 11.45 & 11.46: Solid Forged Bracket Structure

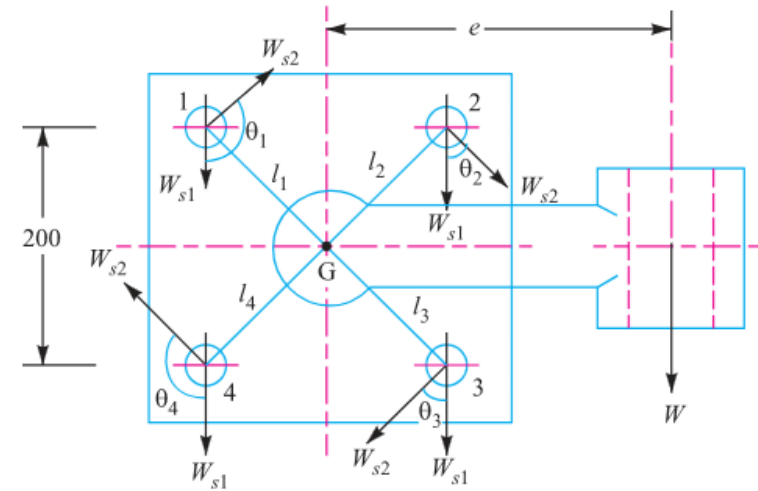


Fig. 11.47

Figure 11.47: Vector Shear Load Diagram on Fastener Group